Sahaj Bhadja

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Summary

Forward-deployed AI engineer with 2+ years building **Python**, **LLM workflows**, retrieval systems, APIs, and production data pipelines while translating ambiguous operational needs into measurable software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Database Systems, Analysis of Algorithms, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Database Management Systems, Cloud Computing, Web Technologies

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Built data integration pipelines using FastAPI, SQL Server, and cloud storage, processing 5K+ monthly jobs with incremental loads, reconciliation checks, error handling, and CI/CD-backed releases at 99%+ reliability.
- Restructured fault-tolerant **LLM orchestration** using FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs while increasing throughput threefold under concurrent workloads with reliable recovery.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Engineered an AI grading platform using multimodal LLMs and OCR to evaluate 10K+ text and handwritten submissions daily with 90%+ consistency across workflows for 150+ instructors at scale.
- Implemented an AWS-hosted backend handling 10K+ daily requests with REST APIs, CI/CD-driven rubric workflows, model-usage tracking, and real-time operator feedback that reduced turnaround time by 60% at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*
- Built a GPT-based Decision Transformer and end-to-end Python pipelines processing 100K+ trajectory logs, improving long-horizon planning by 18% through repeatable experiments and closed-loop evaluation at research scale.
- Architected a spatial-temporal encoder integrating HD-map geometry and agent histories through multi-head attention, improving closed-loop planning stability by 15% while validating behavior across simulation scenarios at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Troubleshot large-scale API integrations using OOP patterns, AWS Glue, S3, and SQL, improving response time by 30%, pipeline efficiency by 68%, and maintainability across production deployments at scale.
- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito while collaborating with Microsoft and GitHub stakeholders across releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Mapped 12+ backend integration points, designed the end-to-end workflow, and presented architecture to the client, resolving ambiguous requirements before implementation across mobile and web experiences and reducing design errors by 25%.
- Developed Flask APIs and Firebase workflows for 50K+ booking records, delivering sub-200-millisecond responses, real-time status synchronization, and webhook-driven automation with approximately 70% less manual handling at scale.

Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, C++, Bash
AI & ML: LLM workflows, multimodal models, retrieval systems, embeddings, PyTorch, Hugging Face Transformers, OpenAI APIs, OCR, prompt engineering
Backend & Data: FastAPI, Flask, REST APIs, microservices, ETL, data cleaning, PostgreSQL, SQL Server, MySQL, Firebase, vector search
Cloud & Deployment: GCP, AWS, Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Linux
Engineering: workflow automation, human-in-the-loop systems, evaluation, monitoring, integration testing, debugging, stakeholder communication, Agile/Scrum, documentation

Projects

OpsBridge | Python, FastAPI, retrieval, human-in-the-loop workflows
- Built and tested an auditable workflow service that normalizes inconsistent schemas, retrieves operational context, escalates low-confidence cases for human review, logs decisions, and measures automation outcomes at scale.

Big-Brain | Live | LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP
- Built semantic retrieval and summarization over 50K+ documents using LLaMA-2, FAISS, Pinecone, and REST APIs, maintaining 95% query accuracy while enabling secure collaborative analysis for 200+ users.

Chat2SQL | Code | Java, TypeScript, SQL, Docker
- Built an AI-assisted SQL workspace supporting 10+ databases, combining schema-aware query generation, structured validation, and Docker deployment to accelerate enterprise data exploration from one unified interface at scale.